Progressive Symmetric Registration for Multimodal Remote Sensing Imagery

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***Abstract*— Image registration forms the foundation of collaborative processing in multimodal remote sensing imagery (MRSI). However, high-resolution MRSIs frequently display complex distortions due to imaging characteristics and terrain variations, with both global and local distortions present. Effectively addressing these complex distortions necessitates the identification of uniformly and densely distributed corresponding points across the entire image. Existing methods primarily focus on global affine distortions and often extract only sparse and unevenly distributed corresponding points, which makes the effective handling of these coexisting distortions a significant challenge. To address this problem, we propose a progressive symmetric registration learning network (PSRNet) for MRSIs. In PSRNet, multimodal remote sensing image registration (MRSIR) is redefined as a symmetric dense regression task, differing from the traditional pipeline that concentrates on unidirectional sparse transformation parameter prediction. Specifically, PSRNet consists of three primary components: 1) a multiscale feature projector (MFP), which employs a dual-branch structure with nonshared weights to achieve modality-specific representation of different modal images across multiple scales, 2) a progressive cross-modal transformer (PCMT) to further mine modality-invariant features and progressively predict symmetric deformation fields, and 3) a symmetric consistency loss (SCL) function capable of elegantly achieving high-precision reversible alignment of image pairs, encompassing endpoint error loss, bidirectional alignment loss, and smoothness loss. Experimental results demonstrate that PSRNet achieves more comprehensive and advanced registration performance on our self-constructed large-scale high-resolution MRSIR dataset, which includes complex global–local geometric distortions and significant nonlinear radiometric differences (NRD).**

***Index Terms*— Deep learning, multimodal registration, remote sensing, symmetric consistency.**

I. Introduction

MULTIMODAL remote sensing data, when synergistically processed and intelligently interpreted, are showing ever-increasing significance in the field of remote sensing. For Earth observation applications like urban land planning [1], environmental monitoring [2], and disaster response [3], combining multimodal sensors is essential to comprehensively observe the scene from different perspectives, including multispectral (MS), hyperspectral (HS), synthetic aperture radar (SAR), light detection and ranging (LiDAR), and thermal infrared (TIR) [4]. Multimodal observations of the same area yield comprehensive information encompassing various wavelengths, resolutions, and viewpoints, enhancing the accuracy of surface matter perception. Monomodal remote sensing data, lacking diverse perspectives and spectral information, often results in incomplete insights and may have limited performance in discriminating features with similar characteristics. Considerable performance improvements have emerged from the collaborative processing of multimodal remote sensing imageries (MRSIs) [5], which enhances the ability to capture intermodal correspondences and provides deeper insights into natural phenomena. MRSIR serves as a fundamental prerequisite and a crucial procedure for integrating valuable and complementary information from heterogeneous modal data. This importance has led to MRSIR becoming an increasingly prominent field of research in recent years.

MRSIR is an image-processing technique that aligns various images of the same scene captured at different times, viewing angles, and with multiple sensors, taking into consideration the significant geometric distortions and nonlinear radiometric difference (NRD) inherent in these images [6]. The primary objective is to achieve pixel-level alignment of the overlapping region, accomplished by geometrically transforming a sensed image into the spatial coordinate system of a reference image using the optimal geometric transformation model. MRSIR methods can generally be categorized into two types: feature-driven methods (FDMs) and data-driven methods (DDMs). FDM establishes a correspondence between image pairs using two approaches: area-based registration [7], [8] and feature-based registration [9], [10]. The area-based registration is founded on a predefined similarity criterion and utilizes an optimal search strategy to iteratively identify transformation model parameters that maximize or minimize the similarity measurement. The feature-based registration formulates correspondences by detecting and matching distinctive abstract features, such as point and line segments. However, these methods consistently confront substantial challenges, primarily arising from the rapid evolution and increasing complexity of multimodal remote sensing imaging technologies, which introduce novel variables and diverse data characteristics. The

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fundamental challenge lies in the limited capacity of these methods to extract modality-invariant advanced representations, primarily focused on low-level features of the imagery. FDM founded on specific hand-crafted features is chiefly suited for particular bimodal challenges, thereby limiting its overall applicability to omnimodal remote sensing imagery. In contrast to FDM, DDM utilizes various nonlinear mapping networks and learnable mechanisms to extract deeper and more abstract modality-invariant features, thereby exhibiting robust registration performance across diverse datasets. DDM typically includes three learning strategies: integrated learning, modality transfer learning, and end-to-end learning [11]. The first learning strategy involves flexibly replacing certain components of traditional registration pipelines with deep neural networks, enabling the construction of comprehensive registration models that integrate various valuable structures according to specific requirements [12], [13]. The second learning strategy simplifies MRSIR to unimodal registration by transforming images from different modalities into a consistent style, bridging the gap between multiple modalities [14], [15]. The third learning strategy utilizes deep neural networks to directly regress sparse or dense geometric transformation parameters, eliminating the need for additional processes such as feature extraction and matching [16], [17].

The primary challenges of MRSIR arise from the complex interactions of different imaging technologies, capture perspectives, and terrain variations, which result in complex geometric distortions and significant NRD. The inaccurate measurements of satellite ephemeris and instrument calibration lead to deviations in the rational polynomial coefficients (RPC) provided by commercial satellite imagery vendors, resulting in georeferenced images exhibiting spatial offsets that typically span several pixels to dozens of pixels in the image space. At present, most research on MRSIR assumes that the geometric correspondence between remote sensing data geo-referenced directly by RPC parameters need only follow an affine model. However, the reality is that RPCs can only minimize significant rotational and scale variations, and a simple affine transformation model is insufficient to accurately describe the complex geometric distortions between MRSIs. These complex geometric distortions are considered to involve both global and local deformations simultaneously. As illustrated in Fig. 1, a pair of optical and SAR images from the WorldView-2 satellite and Capella Space exhibit severe NRD and concurrent global-local geometric distortions. The global geometric distortion between MRSIs is mainly caused by sensor characteristics and the Earth's rotation, which appears as the overall offset, rotation, and scaling of the image. In contrast, the local geometric distortion between MRSIs originates from terrain undulations and capture perspective, etc., and mainly manifests itself as stretching or compression of specific areas.

For the global–local geometric distortion problem, it is necessary to explore the dense registration strategy so as to find the displacement vector at each pixel position. This is mostly implemented by optical flow techniques in the computer vision field, which are well-suited for visible images, but their application to MRSIs is difficult due to the large multimodal

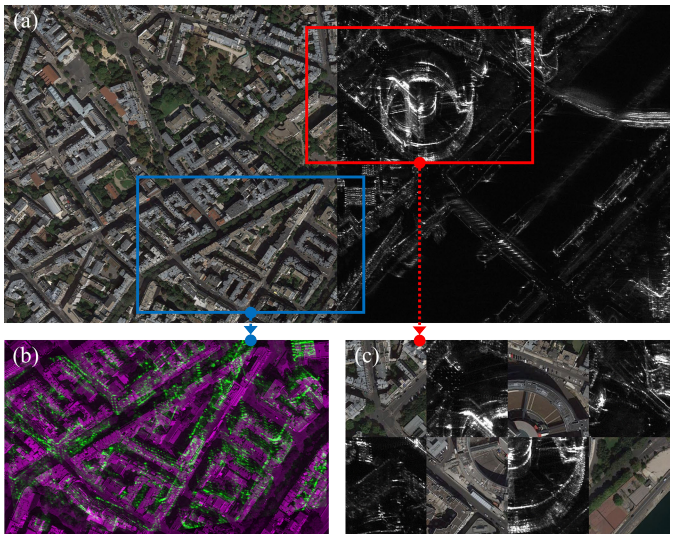


Fig. 1. MRSIs geo-referenced directly by RPCs. (a) Corresponding optical and SAR images from the WorldView-2 satellite and Capella space. (b) Blended image shows the apparent ghosting caused by spatial misalignment, where magenta and green represent optical and SAR content. (c) Checkerboard image reveals an unnatural connection of the same building between adjacent blocks.

gap. In this article, we propose a progressive symmetric registration method for MRSIs, which directly learns the geometric mapping between MRSIs in an end-to-end manner. MRSIR is redefined as a symmetric dense regression task, capable of simultaneously predicting global–local deformation fields. Our method incorporates three principal components: a multiscale feature projector (MFP), a progressive cross-modal transformer (PCMT), and a symmetric consistency loss (SCL) function. The MFP module extracts multiscale modality-specific representations through a dual-branch architecture without shared parameters, while our PCMT module learns modality-invariant features and progressively regresses to precise bidirectional deformation fields. To further refine the deformation fields, the loss function specifically focuses on enhancing the reversibility and accuracy of the deformation fields. The main contributions of this article are summarized as follows.

1) A novel end-to-end progressive symmetric registration method, namely progressive symmetric registration learning network (PSRNet), is proposed for MRSIs. Our method adeptly tackles significant NRD and concurrent global–local distortions by transforming traditional single-scale, unidirectional registration into an advanced multiscale, progressive, symmetric dense deformation field prediction.
2) Two key modules, the MFP and the PCMT, have been designed to encode modality-specific representations at multiple scales and extract common modality-invariant features, enabling stable prediction of dense deformation fields.
3) The symmetric-consistency loss function, specifically designed for symmetric image registration, imposes reversible constraints on a pair of deformation fields while preventing excessive deformation discrepancies

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between neighboring positions, thereby ensuring more accurate and smoother deformation fields.

The rest of this article is arranged as follows. Section II discusses the related works of MRSIR. Section III introduces the proposed PSRNet and its technical details. Section IV details the high-resolution multimodal remote sensing image registration (MRSIR) dataset, along with comprehensive experiments conducted to evaluate our proposed method. Section V provides a summary.

II. Related Works

MRSIR methods are generally classified into two categories: feature-driven registration methods and data-driven registration methods. This section reviews the related work in these categories.

A. Feature-Driven Registration Methods

FDMs align two images through a classic pipeline that involves feature extraction, correct correspondence identification, and transformation model parameter prediction. The primary focus of these methods lies in the development of robust descriptors and the accurate determination of correspondences. The scale-invariant feature transform (SIFT) [18], a pioneering work in this field, is noted for its feature descriptors that remain invariant under translation, rotation, and scale variations. Inspired by SIFT, various improvements have been developed specifically for remote sensing imagery, such as UR-SIFT [19], SAR-SIFT [20], and OS-SIFT [21]. Nevertheless, when confronted with MRSIs, these methods experience noticeable performance deterioration as they struggle to tackle a multitude of challenges including radiation, geometry, scale, viewpoint, dimensionality, and temporal variations.

Recent significant advancements have been made in research on feature-driven MRSIR [22], [23]. Li et al. [24] propose a radiation variation insensitive feature transform (RIFT) algorithm, introducing a maximal index map in lieu of gradients to describe local features. This algorithm aims to address the significant challenges associated with the repeatability and uniqueness of features between MRSIs, ensuring more correct correspondences. Furthermore, a scale and rotation invariant feature transform (SRIF) [25] method with lower complexity has been proposed. This method achieves scale invariance of key points by utilizing a simple pyramid scale space instead of a Gaussian pyramid, and enhances structural information of MRSIs via local intensity binary transformation, thus achieving insensitivity to NRD. Moreover, some studies [26], [27], [28], [29] have indicated that features constructed based on multiscale and multiorientation filtering results exhibit strong robustness to NRD, and combining a pyramid cross-matching strategy contributes to identifying richer pairs of corresponding feature points.

Besides, some researchers build MRSIs' feature representation in a common feature space, thus avoiding the design of multimodal similarity measures and employing classic unimodal similarity metrics like the normalized cross correlation (NCC) to evaluate the intermediate features. Ye et al. [30], [31] design different template matching solutions based on geometric structure properties between MRSIs by combining NCC measure with some specific feature descriptors. These methods employ relatively sparse sampling grids to generate feature descriptors. Consequently, this sparse representation may lead to the neglect of certain available and crucial structural information, thereby compromising the robustness of these methods. To address this limitation, a series of methods based on pixel-wise structural features have been proposed successively. CFOGs [32], OS-PC [33], and SFOC [34] are all highly representative approaches. The shared concept among these methods is to map images from grayscale space to high-dimensional feature space by constructing descriptors, thus capturing more distinctive structural features.

B. Data-Driven Registration Methods

Significant advancements in artificial intelligence within industrial settings have intensified the focus on deep learning research. Deep learning methods, to a certain extent, can address the limitations inherent in traditional approaches and have been effectively incorporated into the field of image registration. Initially, DDMs primarily focused on embedding deep networks into traditional registration frameworks, as detailed in Section I. By training (pseudo) Siamese networks, uniformly distributed and highly repeatable feature points [12], [35] or more discriminative feature descriptors [36], [37], or radiation difference robust feature matches [13], [38] are obtained, enabling the identification of a large number of high-quality correspondences. SuperPoint [12] and Superglue [13] serve as exemplary methods of the integrated learning strategy. They significantly enhance feature repeatability and matching accuracy through the replacement of feature extraction and matching modules in traditional registration methods. Ma et al. [39] transform outlier rejection into a binary classification problem, learning a universal classifier to determine the correctness of any putative match. Furthermore, some studies have explored the possibility of replacing all components of the traditional registration pipeline with trainable and learnable integrated deep networks [38], [40]. Dusmanu et al. [41] employ a single convolutional neural network to simultaneously perform feature detection and extract dense feature descriptors, placing feature detection in a later stage of processing for more stable keypoints. However, the effectiveness of these methods based on an integrated learning strategy is often limited by the performance of other traditional components, leading to error accumulation. Additionally, when processing MRSIs, these methods often struggle to capture sufficiently uniform and dense corresponding points, making it difficult to effectively address the coexistence of global–local geometric distortions.

The modality transfer learning strategy offers a feasible solution to the "multimodal gap" challenge in MRSIR field [14], [42], [43] by transforming images from different modalities into a unified modality. Its core concept involves utilizing generative adversarial networks (GANs) to achieve modality unification, thereby enhancing the performance of traditional unimodal registration methods when applied to MRSIs. Through adversarial training, the generated samples

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are guided to align with the distribution of real data, thereby ensuring that MRSIs exhibit similar radiometric characteristics. For example, Quan et al. [44] propose a GMN method, which utilizes a GAN network to generate simulated optical images corresponding to real SAR images or simulated SAR images corresponding to real optical images. These generated patch pairs are then inputted into a deep matching network, leveraging the latent and correlated features between multimodal patch pairs to infer correspondences. However, patch-based DDM methods tend to perform poorly when there are significant scale and rotation changes between MRSIs. Arar et al. [15] employ spatial transformation network and translation network to achieve MRSIR, not only overcoming the aforementioned limitations but also bypassing the difficulties of developing cross-modal similarity measures. This method encourages the translation network to retain the geometric properties of the images, enabling precise spatial transformation networks to be effectively trained using reliable mono-domain measures. But these methods based on a modality transfer learning strategy may result in the loss of structural details in the original images, particularly in areas with high contrast or rich detail, where important features are prone to being smoothed out or overlooked. Additionally, the translation process can introduce noise or distortion, and even generate nonphysical or unrealistic features, which severely degrade image quality and, in turn, negatively impact the accuracy and reliability of subsequent registration.

The aforementioned methods typically rely on incorporating outlier rejection and transformation model parameter prediction, similar to traditional registration pipelines, to optimize registration results, which prevents them from being true end-to-end deep-learning registration frameworks. Recent research has increasingly shifted toward directly predicting sparse transformation matrices or dense deformation fields, especially in the domains of computer vision and medical imaging [45], [46]. Excitingly, several studies have successfully introduced this idea into the MRSIR field. Zhao et al. [16] develop a deep Lucas–Kanade feature map (DLKFM) tailored for MRSI pairs, extending the traditional Lucas–Kanade algorithm to autonomously identify invariant features between optical images and maps, and employing a Siamese network to predict homography matrix. Li et al. [47] formulate MRSIR as a function of eight displacement parameters that map input reference and sensed images to specified matching points, with the projective transformation parameters calculated via the neural network. Furthermore, Ye et al. [48] explore an unsupervised end-to-end MRSIR method called MU-Net, which is a coarse-to-fine registration method that employs cascaded deep neural networks to directly predict affine transformation model parameters. It utilizes the CFOG descriptor to formulate the loss function for unsupervised training. These methods based on end-to-end learning strategy can directly learn sparse or dense spatial mappings through deep networks, thereby reducing the risk of error accumulation and avoiding the introduction of unrealistic noise.

As the spatial resolution of remote sensing images continues to increase, even slight changes in elevation may cause significant local geometric distortions. Therefore, some researchers are concentrating on developing methods that can tackle both global and local distortions concurrently [49], [50], [51]. Contrary to global registration, which is limited to addressing translation, rotation, and scale, global–local registration engages in a sophisticated image registration procedure by formulating correspondences at a pixel granularity. This procedure accurately aligns images experiencing complex transformations, transcending the constraints of global transformation models like affine or homography [52]. Dosovitskiy et al. [53] first use CNNs for end-to-end dense deformation field prediction, introducing two distinct model variants: FlowNetS and FlowNetC. FlowNetS operates by concatenating two images targeted for deformation field prediction and feeding them into the network. In contrast, FlowNetC processes the images independently through several layers with shared weights to extract features, which are subsequently matched in the correlation layers before being passed through additional layers to complete deformation field prediction. Melekhov et al. [54] propose a dense geometric correspondence network (DGCNet) that employs a hierarchical CNN structure and a coarse-to-fine strategy to capture pixel-wise correspondences, with a matching branch generating probability maps that assign confidence scores to each pixel in predicted deformation field.

III. Methodology

A. Problem Formulation

Given a pair of cross-modal remote sensing images, $I_{\mathrm{A}} \in \mathbb{R}^{H\times W\times 3}$ and $I_{\mathrm{B}} \in \mathbb{R}^{H\times W\times 3}$, each defined in different spatial domains, the challenge is to symmetrically determine the pixel-wise spatial correspondence between them. Symmetrical registration of I_{A} and I_{B} entails establishing an inverse-consistent spatial correspondence, ensuring that the dense deformation field from I_{A} to I_{B}, denoted as $\Phi_f \in \mathbb{R}^{H\times W\times 2}$, and the corresponding field from I_{B} to I_{A}, denoted as $\Phi_b \in \mathbb{R}^{H\times W\times 2}$, are mutually inverse. This consistency harmonizes the complex interaction between the two images, enabling seamless and reciprocal translation. The process of symmetrical registration is elegantly encapsulated in the following formula:

$$I_{\mathrm{A}} = \mathcal{W}(I_{\mathrm{B}}, \Phi_b),\ I_{\mathrm{B}} = \mathcal{W}\left(I_{\mathrm{A}}, \Phi_f\right) \tag{1}$$

where $\mathcal{W}$ means a image warp operation.

In forward registration, each element $\Phi_f[u, v] = [\triangle x, \triangle y] \in \mathbb{R}^2$ represents the deformation offset for a pixel in the unaligned image I_{A}. The registered image $I_{\mathrm{A}}^{\mathrm{reg}}$ is obtained by warping the unaligned image I_{A} using the predicted deformation field Φ_f. The above transformation process can be succinctly described as follows:

$$I_{\mathrm{A}}^{\mathrm{reg}}[u, v] = I_{\mathrm{A}}\left[u + \Delta x, v + \Delta y\right] = I_{\mathrm{B}}[u, v]. \tag{2}$$

From a deep learning perspective, existing methods for dense deformation field prediction employ a resampling strategy to acquire backward deformation fields. However, the resampling strategy inherent in these methods has limitations in handling various scenarios and may introduce noise, as governed by the Nyquist sampling theorem [55]. Here, we propose

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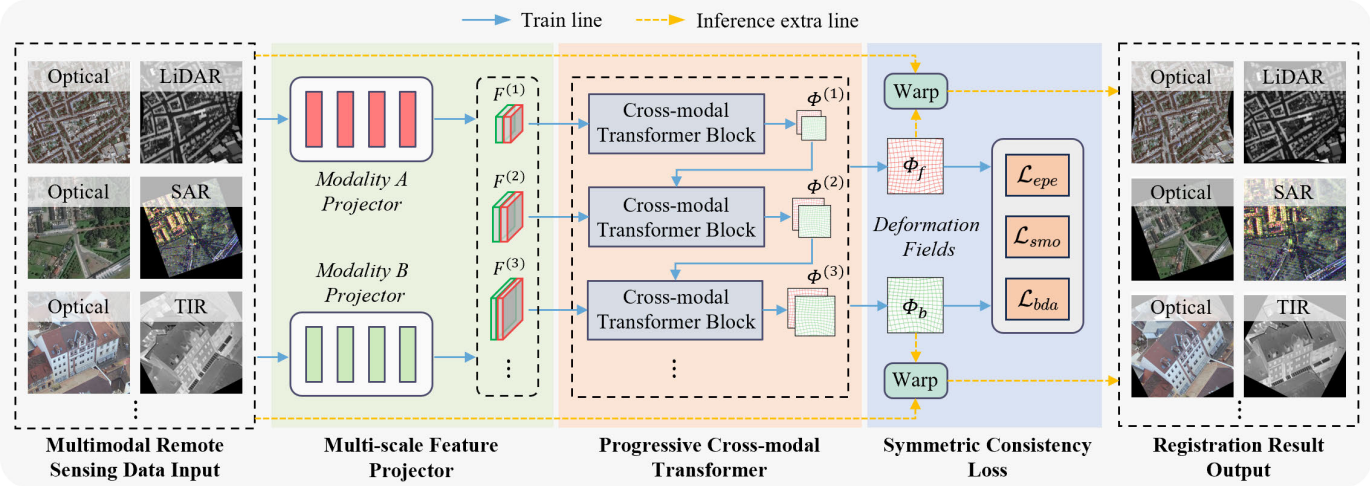


Fig. 2. PSRNet framework consists of 1) MFP, generating multiscale modality-specific representations for each modality using a dual-branch structure with independent weights, 2) PCMT, layering cross-modal transformer blocks to progressively mine modality-invariant features and predict symmetric deformation fields with residuals, and 3) SCL, enforcing reversible alignment in symmetric deformation fields and combining $\mathcal{L}_{\text{epe}}$, $\mathcal{L}_{\text{smo}}$, $\mathcal{L}_{\text{bda}}$ losses.

a symmetric registration network for MRSIs that employs mutual correction between forward and backward registration to prevent unidirectional bias, such as the accumulation of errors in local regions, thereby enhancing robustness to various types of deformations and improving the consistency of forward and backward transformations.

With the above formulated multimodal registration paradigm, PSRNet is designed to achieve symmetric registration of MRSIs through its overall pipeline shown in Fig. 2. PSRNet consists of three main components: an MFP, a PCMT, and an SCL. These modules synergistically complement each other, collectively enhancing the efficiency of registration. For detailed information, refer to the ablation study in Section IV-D. During the inference stage, forward and backward registration results can be obtained by warping the input image using a pair of predicted dense deformation fields.

B. Multiscale Feature Projector

Considering the deep feature maps at different scales is crucial for the MRSIR task. The global structural and shape information in low-resolution feature maps facilitates the correction of overall geometric transformations between images, while the detailed local information in high-resolution feature maps enables the alignment of subtle differences within the images. When conducting image registration solely at a single scale, optimizing dense correspondences becomes challenging and inaccurate. To tackle this issue, we design MFP for feature extraction across multiple scales, capturing various feature changes from global to local levels comprehensively representing the differences between image pairs. Unlike existing methods that rely on CNNs with different convolution kernel sizes for multiscale feature extraction, we employ the pretrained Twins-SVT by ImageNet-1K as the feature projector's backbone and fine-tune it. The Twins-SVT architecture, adeptly integrating local self-attention with global downsampling attention mechanisms, proves eminently suitable for tasks involving geometric registration. It adeptly captures not only minute details but also the overarching structural layout of images. Through the fine-tuning of this pretrained model, its capabilities for multiscale perception are harnessed to their fullest, thereby enabling the extraction of advanced features that are pivotal for effective geometric registration. Given a MRSI pair, I_{A} and I_{B}, MFP employs a dual-branch network with nonshared parameters to extract multiscale features. These branches are referred to as the modality A projector and the modality B projector, respectively. This design with nonshared parameters allows each branch to independently focus on learning and extracting the most valuable features specific to its modality, avoiding interference between modalities. Specifically, each branch uses the first n stages of Twins-SVT [56] as a feature projector, and each stage outputs a $128k$-channel feature map of $(1/2^{k+1})(k \in [1, n])$ image size. The outputs of all stages of each branch network form a feature pyramid F, where the kth level is represented as $F_*^{(k)}$.

C. Progressive Cross-Modal Transformer

Our PCMT module is specifically designed for symmetric deformation field prediction. It processes a pair of feature pyramids F_{A} and F_{B}, progressively predicting the symmetric dense deformation field from the low-resolution feature F_*^1 to the higher resolution feature F_*^n. Each layer of the multiscale features is stepwise fed into the cross-modal transformer to exchange modal invariant information and refine the predictions of the deformation field at different scales, since only relying on the low-resolution feature map to compute the similarity may be unreliable for cross-modal images. As shown in Fig. 3, our PCMT module is composed of multiple cascaded cross-modal transformer blocks, denoted as $\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_n$. The varying sensitivities of features at different scales to global and local distortions necessitate unique parameters for each scale in cross-modal transformation blocks. Employing shared parameters across different blocks might result in excessive smoothing of critical features at certain scales, obscuring essential modality-invariant features necessary for

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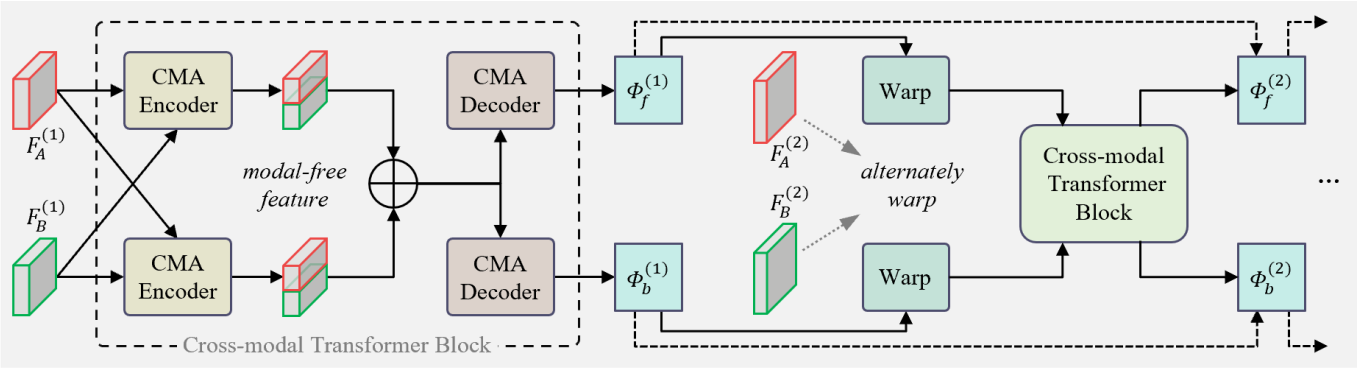


Fig. 3. PCMT schematic.

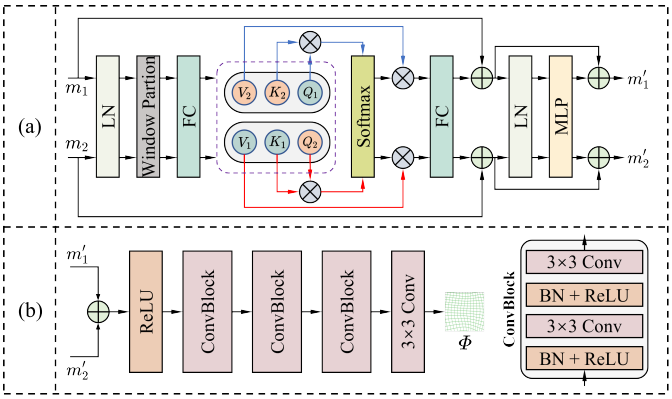


Fig. 4. Cross-modal transformer block. (a) CMA encoder. (b) CMA decoder.

accurate deformation field prediction. Therefore, each cross-modal transformer block possesses independent weights and is specifically tasked with refining the deformation field at its designated scale.

In the kth cross-modal transformer block, the feature map F_*^k is warped by the deformation field $\Phi_*^{(k-1)}$ obtained from the preceding cross-modal transformer block. Let WF and $\Gamma(\cdot)$ denote the warped feature map and upsampling operation, respectively; this can be represented as $WF_*^{(k)} = \mathcal{W}(F_*^{(k)}, \Gamma(\Phi_*^{(k-1)}))$. Then, cross-modal transformer block $\mathcal{T}_k (k \in [2, n])$ alternately takes the $(WF_{\mathrm{A}}^{(k)}, F_{\mathrm{B}}^{(k)})$ and $(F_{\mathrm{A}}^{(k)}, WF_{\mathrm{B}}^{(k)})$ as inputs, predicting corresponding deformation field residuals $\Phi_f^{(k)}$ and $\Phi_b^{(k)}$. Finally, the symmetric deformation fields Φ_f and Φ_b are updated by accumulating the outputs at each layer, ensuring that they maintain the input image scale. This entire progressive process can be mathematically represented as follows:

$$\Phi_f = \Phi_f^{(1)} + \sum_{k=2}^{n} \mathcal{T}_k \left(WF_{\mathrm{A}}^{(k)}, F_{\mathrm{B}}^{(k)} \right) \tag{3}$$

$$\Phi_b = \Phi_b^{(1)} + \sum_{k=2}^{n} \mathcal{T}_k \left(WF_{\mathrm{B}}^{(k)}, F_{\mathrm{A}}^{(k)} \right). \tag{4}$$

Specifically, the cross-modal transformer blocks employ an encoder–decoder architecture, as shown in Fig. 4. The encoder component resembles the Swin Transformer, yet it substitutes the self-attention mechanism with cross-modal attention (CMA). Moreover, the decoder component is composed of BN, ReLU activation, and convolutional layers. This selection is motivated by the inductive bias provided by convolutional neural networks, which is often lacking in transformers.

The CMA encoder block receives a pair of image feature maps extracted by MFP as input for computing attention. Unlike calculating self-attention among different windows within a single feature map, our CMA encoder block enables sufficient communication between features from different modalities to calculate their mutual relevance. As shown in Fig. 4(a), the input features m_1 and m_2 are first partitioned into two sets of windows in the same manner, which aids the subsequent window-based attention computation. These windows alternate between roles as search windows and base windows are projected onto the query set Q_* via a fully connected (FC) layer, and the search windows are projected onto the key set K_* and value set V_*. Next, the CMA calculation occurs between the two sets of windows. In particular, the dot product is computed between the Q_* and K_*, for instance, (Q_1, K_2) and (Q_2, K_1), followed by the application of the softmax activation layer to derive the weights of the values V_1 and V_2. Therefore, the CMA is calculated as follows:

$$\mathrm{CMA}(Q_1, K_2, V_2) = \mathrm{softmax}\left(\frac{Q_1 K_2^{\mathsf{T}}}{\sqrt{d}} \right) V_2 \tag{5}$$

$$\mathrm{CMA}(Q_2, K_1, V_1) = \mathrm{softmax}\left(\frac{Q_2 K_1^{\mathsf{T}}}{\sqrt{d}} \right) V_1. \tag{6}$$

The attention is then projected using an FC and integrated into the base windows via a skip connection. This allows each base window to obtain the relevant weighted information from the search windows, thus effectively establishing remote spatial correspondences between them. Finally, the updated output set is fed into an MLP layer with GELU nonlinearity to enhance the learning capability. LayerNorm (LN) layers are applied before the window partition module to guarantee the effectiveness of each layer.

In the decoding stage, task-specific information is summarized from the CMA features. Existing methods often rely on deep neural network structures to automatically learn complex features from the input. With the powerful support of the CMA encoder block in capturing spatial correspondences between the cross-modal feature inputs, it is efficient to merge the two CMA features and derive the deformation field using a lightweight convolutional neural network as a decoder. As shown in Fig. 4(b), the CMA decoder concatenates the

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CMA encoder's outputs m_1' and m_2' and feeds them into three consecutive convolutional blocks following a ReLU activation layer. Each convolutional block consists of two convolutional layers with a kernel size of 3×3, and before each layer, a batch normalization (BN) layer and a ReLU layer are utilized to enhance the nonlinear feature representation. Finally, a single convolutional layer with a kernel size of 3×3 is used to infer a dense deformation field for registration purposes.

D. SCL Function

Our PSRNet is trained using a supervised learning strategy that jointly predicts bidirectional deformation fields through the symmetric computation of all associated loss terms (i.e., calculating losses from both forward and backward directions). The symmetry constraint we adopt considers the registration of image I_{A} to image I_{B} and image I_{B} to image I_{A} as two independent processes rather than a cyclic process since the cyclic symmetry registration effect depends heavily on the accuracy of the forward registration [51]. The primary loss is the endpoint-error loss $\mathcal{L}_{\mathrm{epe}}$, which is achieved by minimizing the difference between the predicted deformation field and the ground-truth (GT) deformation field. Additionally, two other loss functions, namely bidirectional alignment loss $\mathcal{L}_{\mathrm{bda}}$ and smoothness loss $\mathcal{L}_{\mathrm{smo}}$, are used to constrain predicted symmetric deformation fields.

In tasks that predict dense deformation fields, endpoint error is extensively used as a crucial metric for accuracy assessment. It quantifies the average Euclidean distance between the predicted deformation field and GT deformation field across all pixel points. The endpoint-error loss $\mathcal{L}_{\mathrm{epe}}$ can be constructed between Φ_* and its GT Φ_*^{GT}

$$\mathcal{L}_{\mathrm{epe}} = \frac{1}{N}\sum_{p\in P}\|\Phi_f(p) - \Phi_f^{\mathrm{GT}}(p)\|_2 + \frac{1}{N}\sum_{p\in P}\|\Phi_b(p) - \Phi_b^{\mathrm{GT}}(p)\|_2 \tag{7}$$

where p represents the pixel position of the deformation field, N denotes the total number of pixels, and $\|\cdot\|_2$ represents the l_2 normalization operation.

Owing to the symmetry (reversibility) property of corresponding pixel positions in the forward and backward deformation fields, the sum of the predicted deformations in both directions should be minimized. In accordance with this principle, the bidirectional alignment loss $\mathcal{L}_{\mathrm{bda}}$, formulated to enforce this constraint, is computed using the following formula:

$$\mathcal{L}_{\mathrm{bda}} = \sum_{p\in P}\varpi\left(\Phi_f(p) + \Phi_b\left(p + \Phi_f(p)\right)\right) + \varpi\left(\Phi_b(p) + \Phi_f\left(p + \Phi_b(p)\right)\right). \tag{8}$$

The deformation field is modulated by implementing a second-order smoothness constraint, which serves to suppress significant disparities in the predicted deformation values between adjacent pixels. The smoothness loss $\mathcal{L}_{\mathrm{smo}}$ encourages more effective regularization and guarantees smoother deformation fields. It is calculated using the following formula:

$$\mathcal{L}_{\mathrm{smo}} = \sum_{p\in P}\sum_{(u,v)\in C(p)}\varpi\left(\Phi_f(u) - 2\Phi_f(p) + \Phi_f(v)\right) + \varpi\left(\Phi_b(u) - 2\Phi_b(p) + \Phi_b(v)\right) \tag{9}$$

where $C(p)$ comprises the horizontal, vertical, and diagonal regions surrounding p (a total of four regions). $\varpi(\cdot)$ represents the spatial smoothness penalty function, which is implemented using a generalized Charbonnier function to penalize the smoothness.

Finally, the overall loss function is expressed as follows:

$$\mathcal{L}_{\mathrm{total}} = \lambda_1\mathcal{L}_{\mathrm{epe}} + \lambda_2\mathcal{L}_{\mathrm{bda}} + \lambda_3\mathcal{L}_{\mathrm{smo}} \tag{10}$$

where λ_1, λ_2, and λ_3 represent the weights for each term, and these hyperparameters are discussed in detail in Section IV-D.

IV. Experiment

This section initially constructs a large-scale high-resolution MRSIR dataset, encompassing five distinct modalities. Subsequently, a detailed exposition of the experimental procedures and settings is provided. This is followed by a comprehensive evaluation of the image registration performance of our PSRNet on our constructed dataset, culminating in the conduct of ablation studies.

A. HMRSIR Dataset

To thoroughly assess the effectiveness of the proposed PSRNet for the MRSIR task, we constructed a large-scale, high-resolution dataset specifically designed for MRSIR, termed the HMRSIR dataset. The HMRSIR dataset, comprising five diverse remote sensing modalities including optical, SAR, LiDAR, TIR, and map, is created to address the critical shortage of registration datasets in the MRSIR field, offering a robust benchmark. Given the inherent difficulty in obtaining GT deformation fields between images, a random sample generation process is introduced to create simulated image pairs with known deformation fields. These simulated images are then used for both training and testing registration methods.

1) Generation of Sample Data: The images are distorted by employing three classical transformation models, specifically affine, homography, and thin-plate spline (TPS) transformations. Fig. 5(a) illustrates our data generation process. Given a pair of pixel-level prealigned images I_{A} and I_{B}, we generate the warped sample I_{B}' by applying deformation fields generated by random affine, homography, and TPS transformation parameters to image I_{B}. This process yields image pair $(I_{\mathrm{A}}, I_{\mathrm{B}}')$, where both A→B and B→A possess corresponding GT dense deformation fields of equal sizes. Specifically, affine and homography transformations are controlled by parameters for rotation, scale, and translation. The scale parameters range from 0.5 to 2.0, the rotation parameters range from $-\pi/6$ to $\pi/6$, and the translation parameters range from −45 to 45. Thin-plate spline transformation is defined using a fixed uniform grid of 3×3 control points, as detailed in [57]. Each control point is independently translated in a random direction, with a maximum translation distance of one-quarter of the image size.

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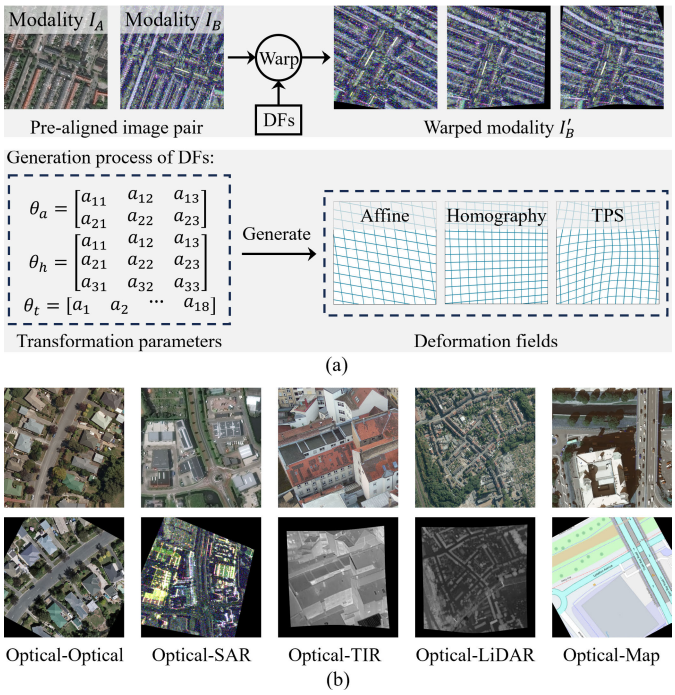


Fig. 5. Illustration of HMRSIR dataset. (a) Generation process of samples. (b) Examples of five cross-modal samples.

2) Dataset: The aforementioned sample generation process substantially augments the diversity of MRSIs. Furthermore, Our HMRSIR dataset provides five combinations of high-resolution cross-modal remote sensing images. The MRSIs in the HMRSIR dataset, marked by significant modal disparities and multitemporal characteristics, are distinguished by pronounced variations in ground targets. The detailed information about each type of combination in the dataset is described as follows.

a) Optical-optical: Two pre-aligned aerial images of the same area in Christchurch, New Zealand, are captured in 2018 and 2020, with a pixel size of 68 989 × 58 954 and an original spatial resolution of 0.075 m. Following the settings outlined in [58], we downsample the spatial resolution of the image pair to 0.3 m and partition the images into three sections according to an 8:1:1 area ratio. Subsequently, we extract 5436, 704, and 978 pairs of patches, each of size 512 × 512 pixels, from the three respective sections, thereby forming the training, validation, and test datasets.

b) Optical-SAR: The 0.5 m spatial resolution polarimetric SAR images and optical images from [59] are selected for sample cropping, covering the area of the Port of Rotterdam in the Netherlands. The size of the image pairs is 37 745 × 29 591 pixels, respectively. Following the same partitioning way as Optical-Optical subdataset, we crop 2183, 272, and 272 pairs of patches for the training, validation, and test datasets, respectively.

c) Optical-TIR: 2360 pairs of RGB and thermal aerial images are chosen from the TBBR dataset [60] to form the Optical-TIR subdataset, with each image measuring 2680 × 3370 pixels in size. These images are captured in 2019 from six large building blocks east of the Market Square in downtown Karlsruhe, Germany. We divide each block according to an 8:1:1 ratio based on the number of images, and crop two nonoverlapping patches of size 1300 × 1300 from each image. Subsequently, these patches are resized to 512 × 512 pixels. The final training, validation, and test datasets comprise 3776, 472, and 472 pairs of patches, respectively.

TABLE I
DETAILED INFORMATION OF HMRSIR DATASET

Modality	Samples			Displacement	
	Train	Valid	Test	Mean	Max
Optical-Optical	16308	2112	2934	17.33	151.97
Optical-SAR	6549	816	816	17.61	136.27
Optical-Thermal	11328	1416	1416	17.40	154.06
Optical-LiDAR	12897	1611	1611	17.46	162.51
Optical-Map	14340	1791	1791	17.41	154.06

d) Optical-LiDAR: The GEONRW dataset [61] consists of 7783 pairs of orthorectified aerial optical images and LiDAR-derived digital elevation models captured in urban areas of North Rhine-Westphalia, Germany, with a spatial resolution of 1 m per image. We stitch all the images into two large images and then crop and select 20 pairs of subimages that cover different areas. Following the same procedure as in Optical-SAR subdataset, we divide each pair of subimage into three sections and crop them, resulting in 4299, 537, and 537 pairs of patches for the training, validation, and testing datasets, respectively.

e) Optical-map: The aerial optical image from 2012 in Optical-Optical and the corresponding map from OpenStreetMap are employed to crop and create the Optical-Map subdataset. Following the same procedure as in Optical-Optical subdataset, we obtained 4780, 597, and 597 pairs of patches for training, validation, and test datasets, respectively.

For a comprehensive assessment of efficacy, each generated patch undergoes three random sample generation processes, employing affine, homography, and TPS transformations, respectively. The detailed characteristics of each combination in the final HMRSI dataset are provided in Table I. Besides the optical images, the pixel values of the other modalities are normalized to the interval $[0, 255]$ to satisfy the input requirements of certain traditional registration methods and to ensure a fair comparison. Notably, during the training process, we implemented various data augmentation techniques on the generated 512 × 512 patches, including random cropping, rotation, horizontal flipping, and vertical flipping. These techniques are designed to significantly enhance the models' generalization ability, resulting in samples of size 256 × 256. During the validation and test process, the patch underwent only center cropping to 256 × 256. Thus, the default size for the patches input to the models is set to 256 × 256. Fig. 5(b) illustrates examples of different subdatasets in the HMRSI dataset, providing readers with a more intuitive understanding.

B. Experimental Settings

1) Evaluation Protocol: The article adopts the standard evaluation metric, average endpoint error (AEPE), for

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predicting dense deformation fields. The endpoint error is determined by calculating the mean Euclidean distance between the predicted deformation field and its corresponding GT, as previously mentioned. The AEPE is then obtained by averaging these endpoint-error values across all images. Additionally, the percentage of correct test (PCT), quantifying the proportion of images with the endpoint error below a specified threshold δ, serves as another criterion for assessing registration performance

$$\text{AEPE} = \frac{\sum_i^N \left(\text{EPE}^i_{\text{A}\to\text{B}} + \text{EPE}^i_{\text{B}\to\text{A}}\right)}{2N} \tag{11}$$

$$\text{PCT}-\delta = \frac{N^\delta_{\text{A}\to\text{B}} + N^\delta_{\text{B}\to\text{A}}}{2N} \tag{12}$$

where N represents the total number of input image pairs, and N^δ indicates the count of test images for which the EPE falls within the predefined threshold δ.

2) Comparative Methods: The registration performance of proposed PSRNet is observed by comparing with the current state-of-the-art methods, including four FDMs (CFOG [32], HOPC [30], CoFSM [26], and RIFT [24]) and four DDMs (FlowNetS [53], FlowNetC [53], PWCNet [49], and DGC-Net [54]), evaluating their adaptability to remote sensing imagery from various sensor types. For all comparative methods, we employ the parameter settings as recommended by the respective authors. It is noteworthy that we leverage GT deformation fields to establish the requisite initial control points for the CFOG and HOPC methods. Each of the four traditional methods derives its corresponding deformation fields from predicted control points, while all four DDMs incorporate data augmentation techniques throughout the training process.

3) Implementation Details: The proposed PSRNet is trained using an AdamW optimizer with an exponentially decaying learning rate initialized to 0.0001 for 200 epochs. The batch size is set to 12, and $\lambda_1, \lambda_2, \lambda_3$, and n are assigned as 0.6, 0.2, 0.2, and 3. Additionally, all resampling operations in our experiments utilize bilinear interpolation. Our method is implemented based on PyTorch, and it is performed on a GPU with NVIDIA RTX 3090.

C. Result Analysis

This section undertakes a detailed comparative analysis between our PSRNet and existing methods, employing the HMRSIR test dataset as elaborated in Section IV-A, and systematically delineates the outcomes of the qualitative and quantitative evaluations.

1) Qualitative Accuracy Analysis: Fig. 6 shows the symmetric registration results for five representative pairs of cross-modal images. For each cross-modal image pair, the first row displays the forward registration result, that is, from image A to B, while the second row shows the backward registration result, namely from image B to A. To ensure fairness in the evaluation of overall registration precision and to mitigate the adverse effects associated with excessively high EPE values, which typically signify registration failure, we implement threshold truncation on the registration results. Specifically, EPE values that exceed the threshold are replaced with the threshold value, which is set at 20 in this work. Across all five cross-modal image pairs, the symmetric registration results of our PSRNet closely resemble their corresponding GTs in both appearance and geometry, highlighting its superior registration performance. However, the comparative methods fail to ensure robustness across all five cross-modal image pairs simultaneously, and there are significant discrepancies in both forward and backward registration performance, indicating certain limitations in their applicability.

Despite the fact that all four traditional registration methods can successfully achieve coarse registration for Optical-Optical image pairs, the CFOG, CoFSM, and RIFT methods exhibit significant misregistration in other cross-modal cases. Specifically, the CoFSM method shows severe displacement when registering optical-SAR, Optical-LiDAR, and Optical-Map image pairs. The primary reason is that its detector can only identify a limited number of control points that are sparsely and nonuniformly distributed in high-resolution image pairs, while also implementing a stringent location-based outlier removal strategy. However, as a bright spot amidst the regret, it is noteworthy that the CoFSM method achieved the second-best registration performance on Optical-TIR image pair. Additionally, CFOG and HOPC methods have also achieved registration accuracy comparable to DDMs on Opitcal-LiDAR image pair.

All four DDMs achieve coarse registration of five cross-modal image pairs. However, each of these methods can only accurately align a limited number of cross-modal image pairs. For example, FlowNetS and FlowNetC, respectively, exhibit good registration performance on Optical-LiDAR and Optical-SAR image pairs, PWCNet achieves the third-best registration accuracy on Optical-TIR image pairs, and DGCNet suffers from severe boundary artifacts on image pairs other than Optical-LiDAR image pair. In summary, a common limitation of all compared methods is the weak registration bidirectional consistency, especially when compared to our PSRNet. The strength of the proposed PSRNet lies in its cross-modal mechanisms, adept at bridging the multimodal gap between images. Additionally, it promotes mutual adjustment for symmetric registration tasks under symmetric constraint conditions.

To more intuitively assess the registration performance of various methods, we selected five representative MRSI pairs from five test datasets for qualitative evaluation. Fig. 7 presents a visualization of the symmetric registration results for MRSIs using our method. The first row corresponds to the input images A and B of two different modalities, the second and third rows correspond to the local deep feature maps extracted by our PSRNet, and the fourth and fifth rows correspond to the checkerboard mosaic images after forward registration (A→B) and backward registration (B→A), respectively. It is evident that our PSRNet extracts highly consistent semantic features across all cross-modal image pairs, which are crucial for progressively optimizing dense deformation fields. Our PSRNet not only accurately aligns buildings and road markings in the Optical-Optical image pair but also effectively connects and aligns small targets, such as windows and minor structures, in the Optical-TIR image pair. Most notably, our PSRNet demonstrates precise alignment in the Optical-SAR image

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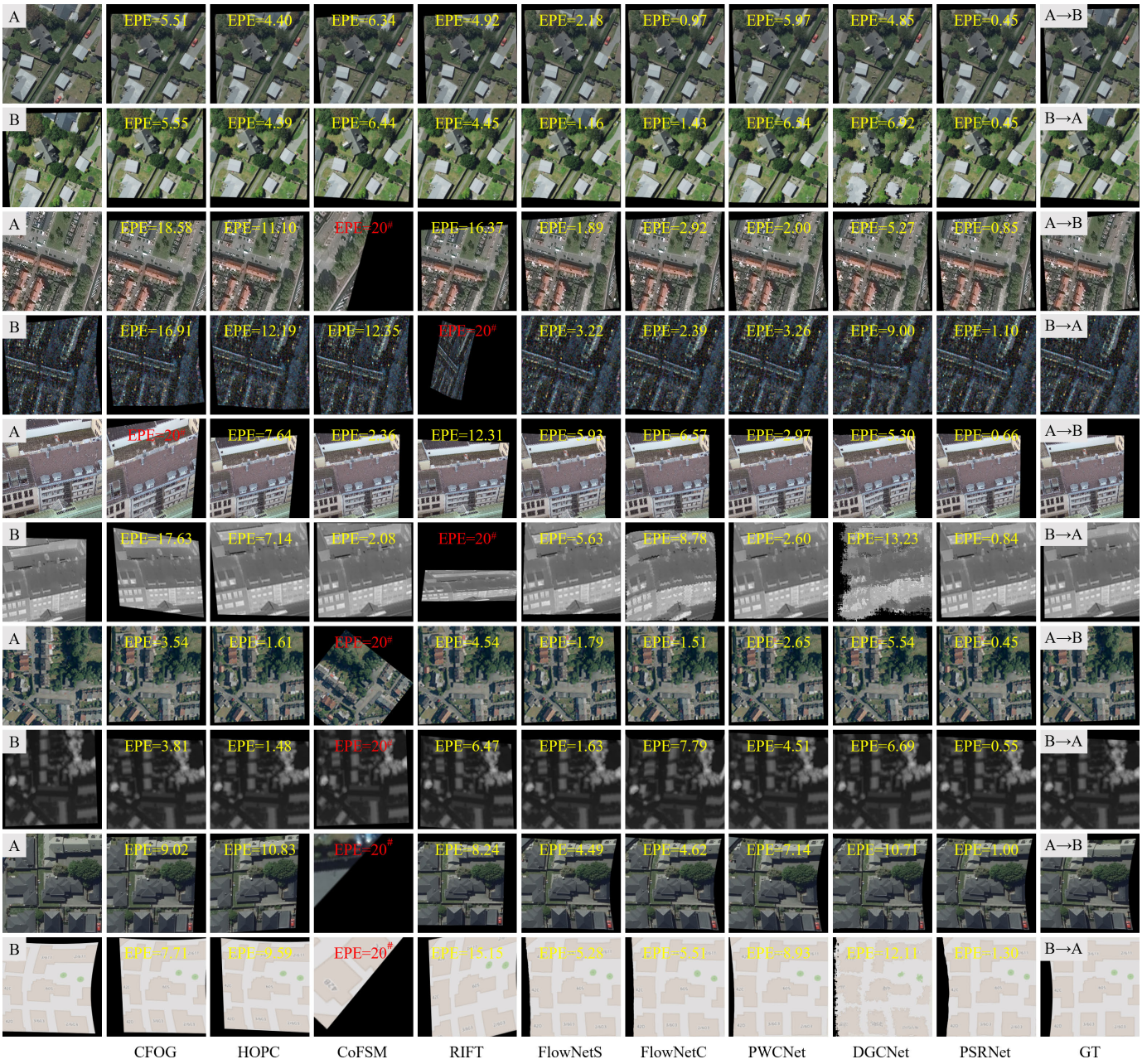


Fig. 6. Visualization of registration results for Optical-optical, Optical-SAR, Optical-TIR, Optical-LiDAR, and Optical-map image pairs. Corresponding EPE values are annotated in yellow font, while registration failures are highlighted in red font.

pair, despite substantial appearance differences and significant speckle noise interference.

2) Quantitative Accuracy Analysis: All methods are evaluated using AEPE and PCT metrics, as listed in Table II. Our PSRNet attains AEPE values of 1.56, 2.18, 2.32, 1.24, and 2.75 across the Optical-Optical, Optical-SAR, Optical-TIR, Optical-LiDAR, and Optical-Map subdatasets, yielding enhancements of 52%, 59%, 60%, 69%, and 56% over the second-best method, respectively. Simultaneously, our PSRNet exhibits outstanding performance on the PCT metric, successfully achieving registration for nearly all MRSIs. This underscores the high reliability of our method, with virtually no instances of failure. Notably, our method accurately aligns 84% of the test samples across all five subdatasets within a more stringent 3-pixel threshold, further illustrating its superiority over competing methods regarding registration robustness and generalizability. Furthermore, our PSRNet achieves the most balanced forward and backward registration accuracy across all methods.

The four feature-driven comparative methods perform poorly across all five subdatasets. Even within the relatively straightforward Optical-Optical subdataset, the proportion of registration results falling below the 3-pixel threshold remains under 10%. Moreover, their AEPE exceeds 13 pixels for Optical-Map and Optical-SAR subdatasets distinguished by marked appearance differences, thereby resulting in lower registration accuracy. This poor performance is likely because these methods only predict the global transformation matrix

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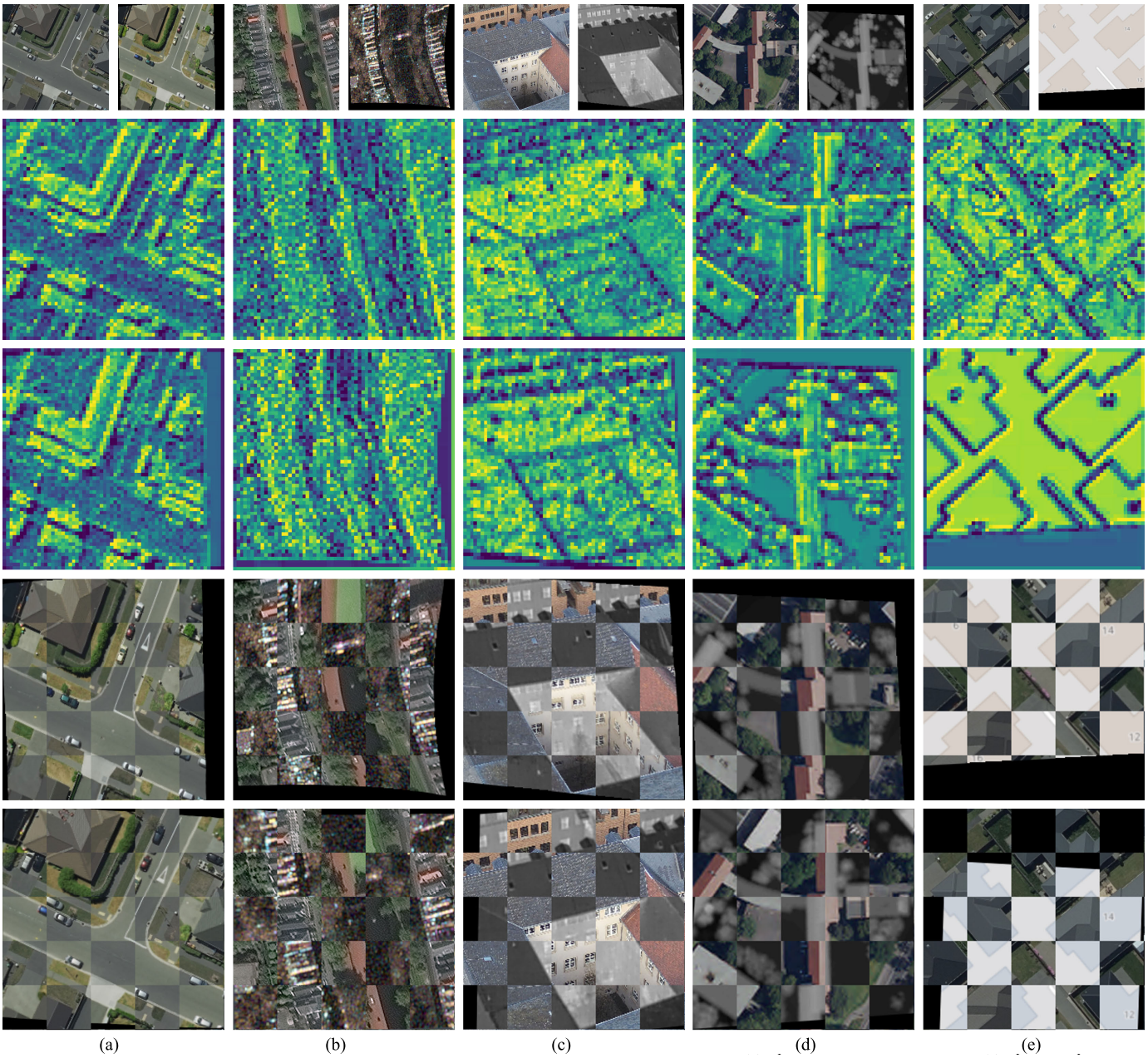


Fig. 7. Visualization of local deep map and checkerboard mosaic map of symmetric registration using PSRNet. (Top bottom) Unaligned image pairs, deep maps of images A and B, checkerboard mosaic maps for forward ($A \rightarrow B$) and backward registration. (a) Optical-Optical. (b) Optical-SAR. (c) Optical-TIR. (d) Optical-LiDAR. (e) Optical-Map.

between images. Moreover, MRSI pairs in our HMRSIR dataset contain local distortions, presenting significant challenges to the comprehensiveness and robustness of these methods. FlowNetS achieves the second-best registration performance, particularly excelling in the Optical-TIR, Optical-LiDAR, and Optical-Map subdatasets. FlowNetC ranks second in registration performance for the remaining two subdatasets. Notably, FlowNetC exhibits a significant imbalance in forward and backward registration performance for the Optical-LiDAR and Optical-TIR subdatasets, with an AEPE value difference exceeding 9 pixels. DGCNet is the worst performer among the five DDMs. It exhibits an average AEPE metric as high as 10.91 pixels across all five cross-modal subdatasets, with registration performance on Optical-LiDAR and Optical-TIR subdataset even lower than FDMs.

3) Evaluation on Different Transformation Types: To evaluate the robustness of all methods under various types of geometric transformations, we meticulously refined the registration results of the methods presented in Table II, categorizing them according to three transformation types: affine, homography, and TPS. The results are illustrated in Fig. 8. Our PSRNet demonstrates a remarkable balance in adaptability across all three types of geometric transformations, consistently achieving optimal registration performance across five subdatasets. The average AEPE across the five subdatasets serves as each transformation model's registration

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TABLE II
QUANTITATIVE COMPARISONS WITH OTHER METHODS, USING AEPE (↓) AND PCT (↑) AS METRICS

Data	Metric		CFOG	HOPC	CoFSM	RIFT	FlowNetS	FlowNetC	PWCNet	DGCNet	PSRNet
Optical-Optical	PCT-3 (%)	A→B	7.17	9.82	3.29	7.22	40.75	58.57	17.58	3.34	**97.25**
		B→A	6.77	10.01	3.19	7.27	38.44	49.93	13.50	0.00	**86.99**
	PCT-5 (%)	A→B	21.99	25.23	15.61	25.72	77.27	89.25	44.72	22.34	**99.51**
		B→A	21.8	25.28	14.92	25.43	73.79	85.52	32.79	0.44	**97.89**
	PCT-10 (%)	A→B	57.98	62.30	55.47	66.86	96.71	99.41	86.11	74.91	**100.00**
		B→A	56.06	60.92	54.15	64.56	93.81	98.58	71.13	34.95	**99.56**
	PCT-20 (%)	A→B	89.54	92.05	85.91	90.92	99.80	**100.00**	99.26	98.04	**100.00**
		B→A	87.97	90.57	84.63	89.74	98.77	**99.90**	92.88	78.74	99.85
	AEPE (pixel)	A→B	9.85	9.32	10.48	8.94	3.97	3.03	6.23	8.09	**1.29**
		B→A	10.06	9.47	10.72	9.15	4.43	3.43	8.15	13.11	**1.83**
Optical-SAR	PCT-3 (%)	A→B	0.12	0.25	0.00	1.72	7.35	13.11	9.19	1.10	**88.73**
		B→A	0.12	0.00	0.00	1.2	6.62	22.55	6.99	0.00	**81.62**
	PCT-5 (%)	A→B	0.49	0.61	0.12	3.06	24.88	41.91	22.06	13.97	**95.71**
		B→A	0.74	0.61	0.12	2.57	30.03	59.93	24.39	1.59	**94.61**
	PCT-10 (%)	A→B	9.31	25.49	2.33	6.86	93.38	93.38	83.95	68.51	**98.90**
		B→A	9.56	25.37	1.96	7.60	79.90	95.34	68.38	31.62	**98.65**
	PCT-20 (%)	A→B	30.02	83.33	13.24	21.69	**100.00**	**100.00**	99.88	97.55	**100.00**
		B→A	30.64	80.64	11.89	23.04	98.53	99.14	97.55	81.50	**99.63**
	AEPE (pixel)	A→B	17.79	13.46	19.18	18.34	6.57	5.80	7.23	8.96	**2.04**
		B→A	17.73	13.60	19.25	18.22	7.28	4.96	8.47	13.01	**2.32**
Optical-TIR	PCT-3 (%)	A→B	10.17	19.56	4.52	6.07	24.29	17.37	12.36	1.41	**91.81**
		B→A	10.10	18.79	4.66	5.16	19.07	0.00	11.58	0.00	**68.50**
	PCT-5 (%)	A→B	24.22	33.55	13.77	17.73	63.91	49.93	40.47	13.21	**96.40**
		B→A	24.51	33.19	14.34	17.16	49.22	0.42	32.91	0.07	**90.61**
	PCT-10 (%)	A→B	50.21	59.82	38.42	42.02	92.02	84.61	84.46	56.00	**98.94**
		B→A	48.80	58.76	36.09	41.45	84.53	15.96	73.31	25.64	**98.23**
	PCT-20 (%)	A→B	72.03	90.89	57.42	63.70	99.29	97.81	98.38	89.76	**99.93**
		B→A	72.46	88.84	56.00	61.72	96.33	58.05	94.63	76.27	**99.79**
	AEPE (pixel)	A→B	11.31	9.08	13.40	12.63	5.15	6.30	6.65	10.45	**1.72**
		B→A	11.36	9.25	13.58	12.88	6.37	15.88	7.92	13.88	**2.91**
Optical-LiDAR	PCT-3 (%)	A→B	13.53	19.89	0.06	1.18	50.84	27.69	39.11	3.97	**98.08**
		B→A	12.85	19.93	0.19	0.87	34.76	0.00	30.85	0.81	**91.00**
	PCT-5 (%)	A→B	27.93	34.02	1.92	6.70	83.12	60.09	73.00	24.33	**99.50**
		B→A	27.68	33.71	1.99	6.08	69.83	0.56	62.38	11.86	**98.39**
	PCT-10 (%)	A→B	54.93	59.22	9.81	30.60	97.95	86.65	95.90	73.99	**99.81**
		B→A	53.94	58.29	9.06	28.49	94.72	21.29	93.48	58.97	**99.75**
	PCT-20 (%)	A→B	81.13	87.83	22.35	55.49	99.94	98.63	99.75	97.21	**100.00**
		B→A	80.26	86.22	21.73	51.96	99.01	69.27	99.19	91.06	**99.88**
	AEPE (pixel)	A→B	10.26	9.27	18.07	14.54	3.52	5.58	4.14	8.15	**0.98**
		B→A	10.36	9.39	18.17	14.87	4.43	14.76	4.85	10.02	**1.49**
Optical-Map	PCT-3 (%)	A→B	3.57	1.95	0.00	5.30	22.78	13.68	11.73	0.50	**75.32**
		B→A	1.90	1.28	0.06	1.06	13.74	13.29	10.72	0.00	**58.24**
	PCT-5 (%)	A→B	9.72	6.03	0.11	7.71	53.60	34.79	27.02	8.93	**93.80**
		B→A	7.20	5.36	0.22	2.96	39.31	36.63	26.30	2.01	**87.16**
	PCT-10 (%)	A→B	29.70	24.23	2.12	21.72	90.51	78.67	69.68	51.65	**99.67**
		B→A	26.19	22.89	2.18	15.75	81.74	80.24	66.11	38.25	**98.94**
	PCT-20 (%)	A→B	53.32	66.22	11.50	42.04	99.67	98.44	98.77	90.23	**100.00**
		B→A	50.14	63.99	11.33	34.06	97.21	98.16	95.53	81.63	**99.94**
	AEPE (pixel)	A→B	14.50	14.37	19.28	15.79	5.52	7.19	8.04	10.85	**2.41**
		B→A	15.05	14.70	19.30	16.93	6.93	7.08	8.63	12.58	**3.08**

accuracy, revealing that the maximum discrepancy in registration accuracy for our PSRNet among the three transformation models remains within 0.8. Notably, the comparative methods perform best under affine transformations, followed by homography transformations, with the weakest performance observed under TPS transformations. This trend arises from the fact that affine and homography transformations involve simpler global distortions, while TPS transformations encompass more complex local distortions. For each transformation mode, the comparative methods show the maximum variation in registration accuracy within the Optical-SAR and Optical-Map subdatasets, while the registration accuracy differences are minimal in the Optical-Optical subdataset. FlowNetS attains a commendable second position in balanced registration performance across the three transformation modes, whereas CFOG and HOPC exhibit the most pronounced fluctuations in registration accuracy among these transformation modes.

4) Evaluation on Robustness for Scale, Rotation, and Translation Distortions: To evaluate the robustness of PSRNet against scale, rotation, and translation distortions, we selected 100 pairs of images from the test dataset of each cross-modal subdataset. These images are subjected to random scale, rotation, and translation operations. Specifically, the scale parameter is confined to a range of $[0.5, 2.0, 0.15]$,

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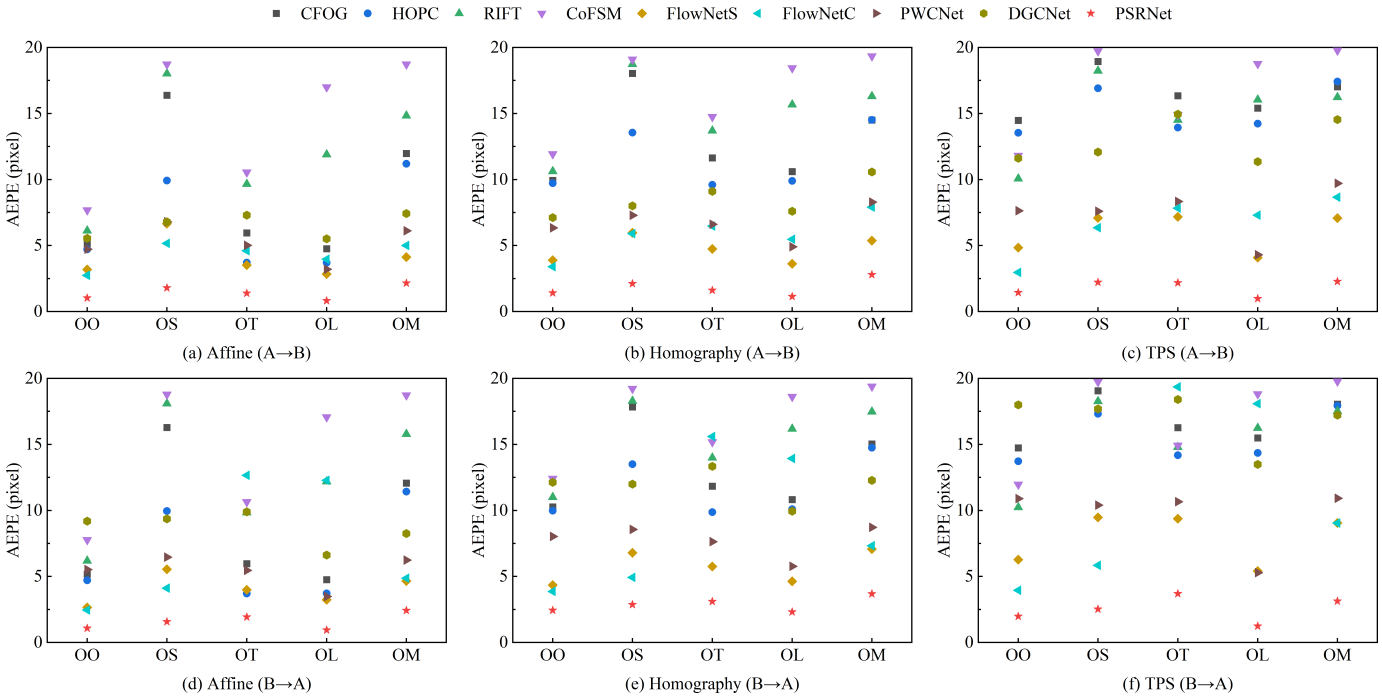


Fig. 8. Comparative evaluation of various methods for MRSIs with affine, homography, and TPS transformations in HMRSIR test dataset. (a) and (d) Affine transformation. (b) and (e) Homography transformation. (c) and (f) TPS transformation. The OO, OS, OT, OL, and OM denote five different subdatasets within our HMRSIR dataset.

TABLE III
COMPLEXITY COMPARISONS OF DIFFERENT METHODS ON THE HMRSIR DATASET, USING TIME (↓) AND FLOPS (↓) AS METRICS

Backend	Metric	CFOG	HOPC	RIFT	CoFSM	FlowNetS	FlowNetC	PWCNet	DGCNet	PSRNet
CPU	**Time(s)**	0.79	2.72	3.18	0.42	**0.27**	0.46	0.49	1.63	1.12
GPU	**Time(s)**	-	-	-	-	0.14	0.13	0.15	0.13	**0.12**
	Params(M)	-	-	-	-	38.68	39.18	**9.37**	10.35	151.01
	FLOPs(G)	-	-	-	-	**8.90**	11.10	12.12	67.03	74.22

with 0.15 representing the step size. The rotation parameter is set within a range of $[-30°, 30°, 6°]$, while the translation parameter is defined within a range of $[-45, 45, 10]$. Fig. 9 illustrates the registration performance of various methods under these distortions for MRSIs.

Our PSRNet exhibits remarkable adaptability to distortions in scale, rotation, and translation, ensuring competitive registration performance even under large-angle rotations and substantial translations. In contrast, the comparative methods display varying degrees of sensitivity to these transformations. While FlowNetS proves robust against rotation and translation changes, it struggles with scale variations. CoFSM exhibits considerable instability under these distortions, and CFOG, HOPC, and DGCNet demonstrate particular sensitivity to scale and rotation distortions, with their AEPE values displaying an inverted pyramid shape that increases with larger scale differences and rotation angles. This signifies that the robustness of these comparative methods regarding scale, rotation, and translation distortions requires considerable enhancement.

5) Complexity Evaluation: The evaluation results of various methods regarding computational complexity on both CPU and GPU are presented in Table III, where each method's results represent the averages across five cross-modal subdatasets. The evaluation metrics under GPU configuration encompass floating-point operations (FLOPs), parameters (Params), and the average registration time per image pair (Time). In contrast, the assessment of registration efficiency under CPU configuration relies solely on the Time metric. Under CPU configuration, FlowNetS exhibits the shortest registration time, while RIFT requires the longest duration, with our PSRNet positioned in between. Under GPU configuration, PSRNet achieves the shortest registration time, but it also has the highest FLOPs and model parameters, indicating greater computational complexity. Nevertheless, given that PSRNet significantly enhances registration accuracy, we consider the increase in computational complexity to be reasonable and acceptable in exchange for improved accuracy.

D. Ablation Study

This section performs a comprehensive ablation study to clarify the influences of the proposed CMA, progressive registration strategy, and SCL on image registration performance. In addition, we analyze the hyperparameters related to the loss functions, the progressive levels of cross-modal transformer blocks, and the simulated data with a black border on model performance.

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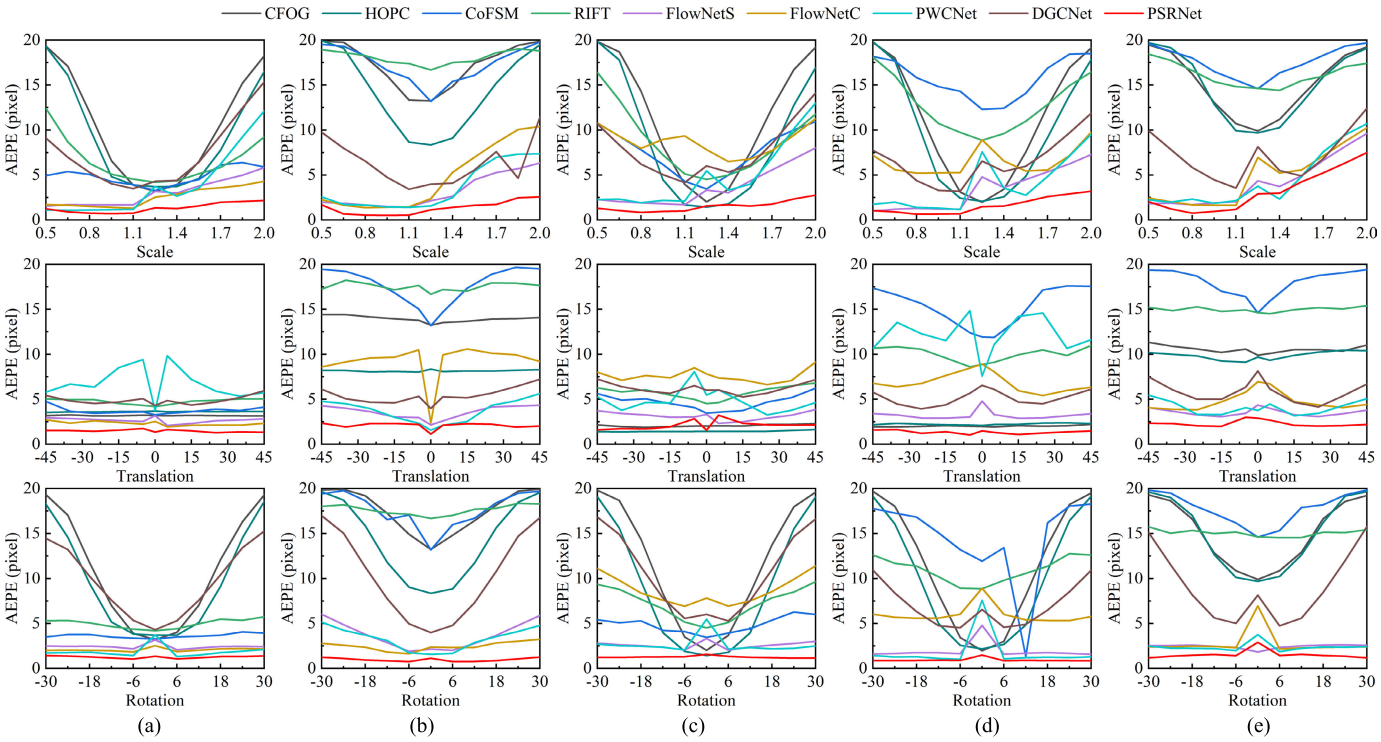


Fig. 9. Comparative evaluation of various methods for MRSIs with scale, translation, and rotation distortions. (a) Optical-Optical. (b) Optical-SAR. (c) Optical-TIR. (d) Optical-LiDAR. (e) Optical-Map.

TABLE IV

QUANTITATIVE COMPARISONS WITH ABLATION MODELS, USING AEPE (↓) AS METRIC

Model	MFP	PCMT	SCL	OO	OS	OT	OL	OM
1	-	-	-	6.90	7.13	8.26	8.12	9.62
2	-	-	✓	6.05	6.39	7.37	7.25	8.72
3	✓	-	✓	4.24	5.01	4.41	4.28	5.83
4	✓	✓	✓	**1.56**	**2.18**	**2.32**	**1.24**	**2.75**

TABLE V

IMPACT OF PROGRESSIVE AND NONSHARED PARAMETER SETTINGS ON REGISTRATION PERFORMANCE IN NETWORK ARCHITECTURES, USING FLOPS (↓), PARAMS (↓), AND AEPE (↓) AS METRIC

Model	FLOPs (G)	Params (M)	AEPE				
			OO	OS	OT	OL	OM
PSRNet-F	**51.96**	101.21	3.66	3.28	4.83	3.31	4.67
PSRNet-S	54.90	**100.54**	2.25	2.72	2.86	2.41	3.13
PSRNet	74.22	151.01	**1.56**	**2.15**	**2.32**	**1.24**	**2.70**

1) Module Analysis: Table IV presents the configurations and quantitative evaluation results of the ablation experiments. "MFP" denotes the multiscale feature projector module, "PCTM" represents the progressive cross-modal transformer module, and "SCL" stands for symmetric consistency loss. Notably, for models without the SCL function, $\mathcal{L}_{\text{epe}}$ is used as the loss function; for models without the PCTM module, ResNet50 is used as the baseline. Experimental results demonstrate that our three key modules improve PSRNet's registration performance to varying degrees. Compared to the backbone network, PSRNet enhances registration accuracy on cross-modal data by 4.95 to 7.14 pixels. Specifically, with the support of our MFP, PCMT, and SCL modules, the average registration accuracy on five cross-modal subdatasets improved by 2.40, 2.74, and 0.85 pixels, respectively.

The impact of progressive and nonshared parameter settings in our PSRNet on the registration performance is further analyzed. As shown in Table V, "PSRNet-F" represents the replacement of the progressive strategy with a multiscale feature fusion strategy. "PSRNet-S" represents the substitution of nonshared parameter configurations in both CMT blocks and MFP with shared parameter configurations. Clearly, compared to the final PSRNet, the PSRNet-F model reduces FLOPs by approximately 30% and parameter count by 33%, but the average registration accuracy on five cross-modal subdatasets decreases by about 98% (2.0 pixels). Similarly, the PSRNet-S model reduces FLOPs by 26% and parameter count by 33%, but the average registration accuracy decreases by about 35% (0.7 pixels). While the progressive approach and nonshared parameter configuration in the proposed method do add to the computational burden and parameter count, we consider these trade-offs worthwhile for the sake of achieving better registration accuracy.

2) Hyperparameter Analysis: We evaluate the impact of different weight combinations for multiple loss functions on registration performance. As shown in Table VI, increasing the weight of λ_1 while decreasing the weights of λ_2 and λ_3 gradually improves the registration accuracy of PSRNet on five subdatasets. However, this improvement is not sustained;

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TABLE VI

EFFECT OF DIFFERENT HYPERPARAMETERS ON REGISTRATION PERFORMANCE, USING AEPE (↓) AS METRIC

$\lambda_1, \lambda_2, \lambda_3$	OO	OS	OT	OL	OM
0.10, 0.45, 0.45	2.47	2.28	2.24	1.60	2.76
0.20, 0.40, 0.40	3.40	2.22	2.12	1.91	2.74
0.40, 0.30, 0.30	1.86	2.33	2.20	1.89	2.74
0.60, 0.20, 0.20	**1.46**	**2.14**	**1.96**	**1.23**	2.71
0.80, 0.10, 0.10	2.15	2.19	2.95	2.18	**2.70**

TABLE VII

EFFECT OF THE NUMBER OF CASCADING CROSS-MODAL TRANSFORMER BLOCK ON REGISTRATION PERFORMANCE, USING AEPE (↓) AND PCT (↑) AS METRIC

Metric	n=1	n=2	n=3	n=4
PCT-3 (%)	38.79	69.55	**85.18**	14.46
PCT-5 (%)	79.17	92.95	**95.16**	57.90
PCT-10 (%)	97.92	**98.84**	98.78	97.73
PCT-20 (%)	99.88	**99.94**	99.82	99.69
AEPE (pixel)	3.91	2.75	**2.18**	4.96

when the weights are set to $(0.8, 0.1, 0.1)$, the registration performance declines on multimodal data, excluding Optical-Map subdataset. The optimal registration performance occurs with weights set to $(0.6, 0.2, 0.2)$. In this setup, $\mathcal{L}_{\text{epe}}$ handles coarse global registration and needs a larger weight, while $\mathcal{L}_{\text{bda}}$ and $\mathcal{L}_{\text{smo}}$ manage fine local registration, requiring smaller weights.

3) Performance Analysis at Progressive Levels: The influence of varying levels of cross-modal transformer blocks on the registration performance of the proposed method is meticulously assessed, similar to existing methods [62], [63]. In particular, a representative Optical-SAR cross-modal subdataset is employed for experimentation. The PSRNet method, implemented with one to four progressive levels, is systematically compared with respect to convergence on the training dataset and registration accuracy on the test dataset. The results are illustrated in Fig. 10 and detailed in Table VII. Fig. 10 shows that as the progressive levels increase, the model converges to lower AEPE values. However, higher progressive levels are not always better. When $n = 4$, the training process of the model exhibits significant fluctuations and fails to converge to better accuracy than $n = 2$ and $n = 3$. The reason is that although the significant differences between cross-modal images require the registration method to have higher-level semantic information for accurate registration, excessive downsampling leads to the loss or blurring of key information, which in turn affects the model's learning and convergence. With $n = 3$ achieving the best convergence and registration performance, it is set as the default number of progressive layers for our PSRNet. Additionally, since GT labels are used to supervise the training of our PSRNet, the cumulative error generated during the progressive process is negligible.

4) Performance Analysis of Black Border: To assess whether the black borders in the simulated image pairs from the HMRSIR dataset affect registration performance,

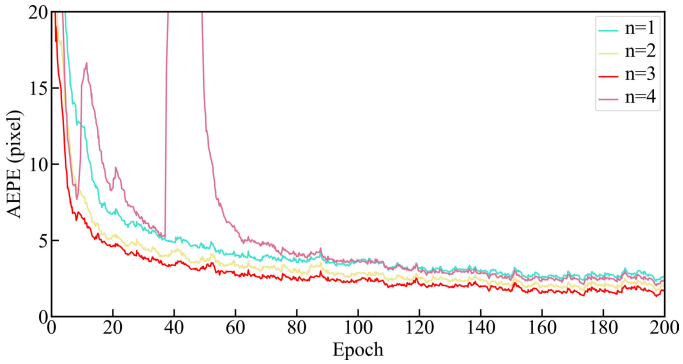


Fig. 10. Effect of varying numbers of cascading cross-modal transformer blocks on model convergence during training.

TABLE VIII

EFFECT OF BLACK BORDER ON REGISTRATION PERFORMANCE, USING AEPE (↓) METRIC

Model	AEPE				
	OO	OS	OT	OL	OM
w/o Black Border	1.59	2.58	2.55	2.02	5.39
w/ Black Border	**1.56**	**2.15**	**2.32**	**1.24**	**2.70**

we replace the black border regions with the border values of the original images and retrain and test the model. As shown in Table VIII, removing the black borders resulted in a decrease in registration performance across the five cross-modal subdatasets. The impact is minimal for the Optical-Optical subdataset, whereas the Opitcal-Map subdataset experiences the most significant decline, with a reduction of approximately 100% (2.7 pixels) in registration accuracy. We consider that image pairs with black borders introduce less interference compared to those without, as the black border pixels have a constant value (0), leading to a simpler distribution. In contrast, the pixel distribution in images without black borders is more complex.

V. CONCLUSION

In this article, a progressive symmetric image registration learning network for MRSIs is proposed. Our PSRNet obtains a pair of symmetric dense deformation fields and enhances the reversibility of registration by focusing on the consistency of forward and backward registration of image pairs. More importantly, a multiscale projector for modality-specific representations and a PCMT module are designed to enhance the capability to learn modality-invariant features from cross-modal image pairs. In addition, by imposing a progressive strategy of gradually refining the dense deformation field parameters from coarse to fine, and the SCL function consisting of multiple loss constraints, the robustness of MRSIs with large displacements is enhanced while maintaining the neat appearance that the images should have. Experiments on a large-scale HMRSIR dataset with five different modality types validate the advantages of our PSRNet. However, the shortcoming of the proposed method is its high-computational complexity compared to other DDMs, which is the direction of our future improvement. In the future, symmetric registration

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for cross-scale MRSIR in a weakly supervised manner will also be the focus of the research.

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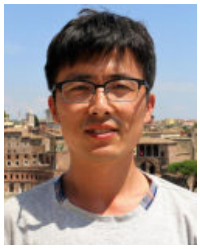
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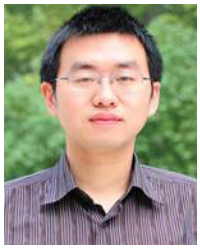
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